

CS2311 Modern Operating System Exercises Spring 2026

Exercise 1: CPU Scheduling

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The following rules apply to the exercises:

* We assume that when a preempted process and new coming process need to be put in the ready queue at the same time, we prefer to put the new coming process in front of the preempted process in the ready queue.

* When processes are of same priority, use FCFS to determine the order.

1. Consider the following set of processes, with the arrival times and the length of the CPU-burst times given in milliseconds, and the priorities given so that smaller priority number means a higher priority.

Process	Arrival time	Burst time	Priority
P1	0	10	3
P2	2	1	1
P3	3	2	3
P4	4	1	4
P5	6	5	2

- a) Draw Gantt charts illustrating the execution of these processes using the following scheduling algorithm: (1) FCFS, (2) nonpreemptive SJF, (3) preemptive SJF, (4) nonpreemptive priority, (5) preemptive priority, (6) RR (quantum = 1), and (7) RR (quantum = 3). We specify that when a preempted process and new coming process need to be put in the ready queue at the same time, we prefer to put the new coming process in front of the preempted process in the ready queue. We also specify that when processes are of the same priority, use FCFS to determine the order.
 - b) Calculate the average waiting time when using each of the above scheduling algorithms.
 - c) Calculate the average turnaround time when using each of the above scheduling algorithms.
2. Consider the following set of processes, with the arrival times and the length of the CPU-burst times given in milliseconds.

Process	Arrival time	Burst time
P1	0	17
P2	12	25
P3	28	8
P4	36	32
P5	46	18

- a) Draw a Gantt chart illustrating the execution of these processes using a Multilevel Feedback Queue Scheduling. Use the same structure of the model as in the lectures: (queue 0: quantum 8), (queue 1: quantum 16), (queue 2: FCFS). If there is a tie, the scheduling algorithm breaks it by preferring the process with a smaller id.
- b) Calculate the average waiting time and the average turnaround time for the scheduling.

3. Consider the following set of processes, with the arrival times, the length of the CPU-burst times, and I/O times given in milliseconds.

Process	Arrival time	Burst time 1	I/O time	Burst time 2
P1	0	32	16	4
P2	10	24	8	6
P3	28	8	8	12
P4	36	12	25	4
P5	44	18	12	8

- a) Draw a Gantt chart illustrating the execution of these processes using a Multilevel Feedback Queue Scheduling. Use the same structure of the model as in the lectures: (queue 0: quantum 8), (queue 1: quantum 16), (queue 2: FCFS). If there is a tie, the scheduling algorithm breaks it by preferring the process with a smaller id.
- b) Calculate the average waiting time and the average turnaround time for the scheduling.
4. Which of the following scheduling algorithms could result in starvation, and why?
- First-come, first-served
 - Shortest job first
 - Round robin
 - Priority